



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO RANAS FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

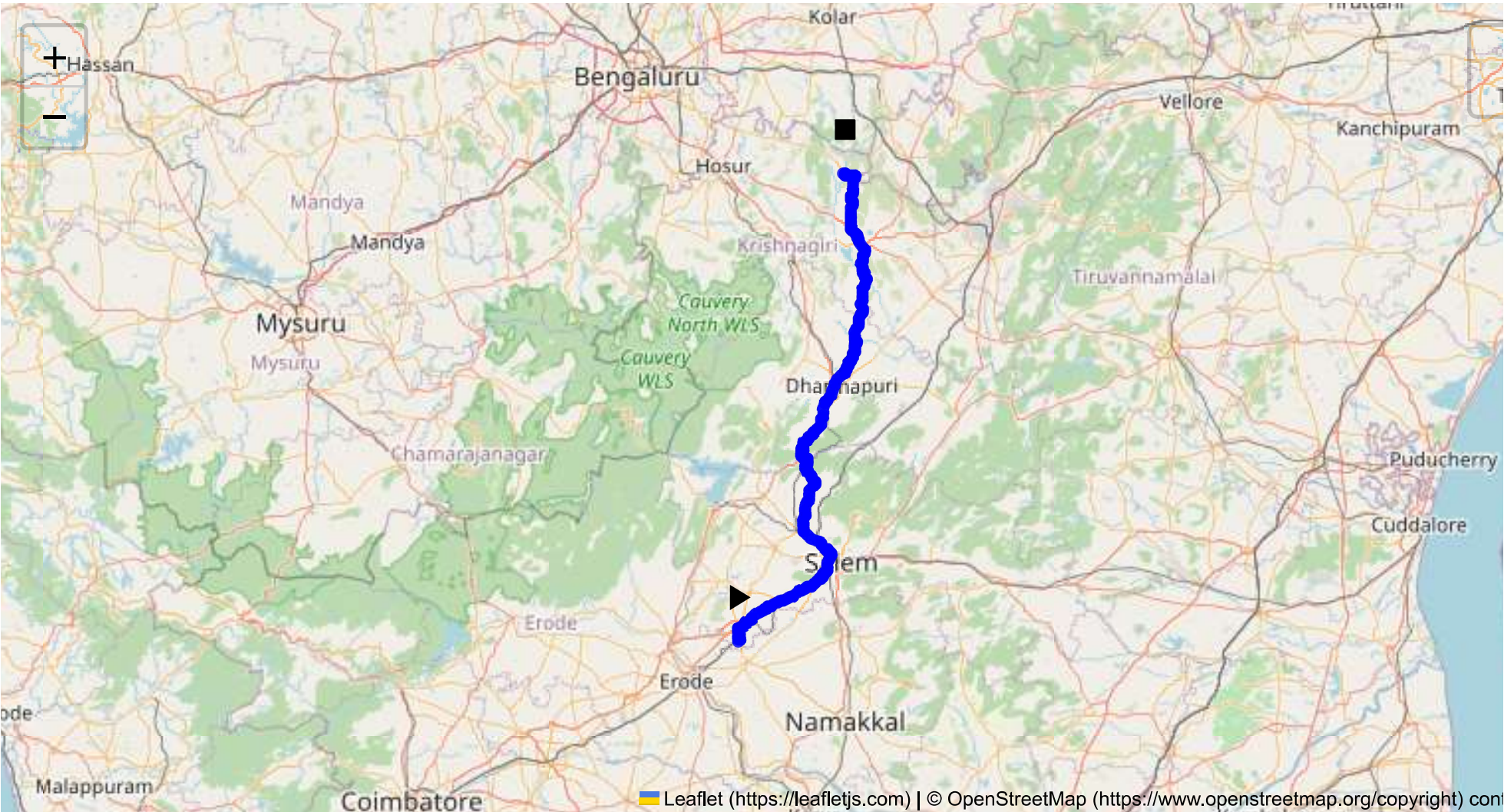
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 177.93 km
Estimated Duration: 3.3 hours
Adjusted Duration (Heavy Vehicle): 4.1 hours
Start: (11.4381, 77.8734)
End: (12.7101, 78.1649)



Welcome to the Journey Risk Management Study

Below is a comprehensive analysis of the route from CVQF+23W, Sangagiri, Tamil Nadu to P567+2X3, Kadavarapalli, Tamil Nadu, intended to help inform and train truck drivers transporting hazardous materials.

1. Overview of the Route Map

The journey spans approximately 177.93 kilometers and travels southeast from Sangagiri to Kadavarapalli. The route either follows National Highways (NH) or State Highways (SH), offering a well-defined pathway with varying road conditions.

2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu experiences a tropical climate with relatively stable temperatures year-round. The monsoon season, which generally occurs from June to September, can bring heavy rainfall, leading to potential flooding and reduced visibility. Drivers should be prepared for slippery roads and reduced traction during and after rains.

3. Analysis of Traffic Patterns

Traffic is moderate, with peak hours generally between 8:00-10:00 AM and 5:00-7:00 PM. Areas around major towns and intersections, such as near Salem and Krishnagiri, are prone to congestion, particularly during weekends and holidays.

4. Assessment of Road Quality and Infrastructure

The road quality is mostly good, with NH and SH maintaining acceptable standards. However, some rural sections might have potholes and narrower lanes. Drivers should be cautious in urban transition zones where sudden changes in road quality can occur.

5. Suggestions for Alternative Routes for Emergencies

In case of emergencies, alternative routes include bypasses around Salem via NH544, which connects to NH44 heading towards Krishnagiri. Additionally, local state highways might be used, although they may have inferior road quality and infrastructure.

6. Summary of Local Regulations Affecting Hazardous Material Transport

Tamil Nadu follows strict guidelines for transporting hazardous materials, including required permits, designated routes during off-peak hours, and mandatory safety equipment. Compliance with the Hazardous Wastes (Management and Handling) Rules is crucial.

7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials

While major incidents are infrequent, there have been occasional reports of truck accidents involving heavy vehicles on NH44 and NH544. Factors often include speeding and loss of control due to adverse weather conditions.

8. Environmental Considerations and Sensitive Areas

The route passes through agricultural zones and near water bodies. It is crucial to prevent spills and respond swiftly in case of a leakage to protect these sensitive environments. Awareness of the proximity to forested regions near Krishnagiri is important.

9. Analysis of Communication Coverage

Mobile network coverage is generally stable throughout the route, with potential dead zones in more rural and forested sections. It’s advisable to have radio communications as a backup.

10. Estimated Emergency Response Times for Different Route Segments

Emergency services in urban areas like Salem and Krishnagiri offer quick response times, typically under 30 minutes. In rural areas, it may extend to an hour due to distance and road conditions.

11. Overall Summary of Risk Assessment

The overall risk assessment indicates a medium level of risk for transporting hazardous materials along this route. Key risks include weather hazards during monsoons, potential congestion in urban zones, and occasional poor road conditions. Preparedness is enhanced by using real-time navigation updates, adhering to regulations, and maintaining clear communication.

In conclusion, while the route is generally secure for transporting hazardous materials, drivers should remain vigilant, especially during inclement weather and in high-traffic areas, and comply with all applicable safety regulations.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43968, 77.87345	15 KM/Hr
1	Turn	High	11.44029, 77.87544	15 KM/Hr
2	Turn	High	11.44900, 77.87406	15 KM/Hr
3	Turn	High	11.45028, 77.87202	15 KM/Hr
4	Turn	Medium	11.46400, 77.87323	30 KM/Hr
5	Turn	Medium	11.46765, 77.87037	30 KM/Hr
6	Turn	High	11.47403, 77.86841	15 KM/Hr
7	Turn	Medium	11.47986, 77.88397	30 KM/Hr
8	Turn	High	11.49149, 77.89601	15 KM/Hr
9	Turn	High	11.49224, 77.89601	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
10	Turn	Medium	11.53512, 77.95717	30 KM/Hr
11	Turn	Medium	11.53606, 77.95869	30 KM/Hr
12	Turn	Medium	11.62403, 78.10893	30 KM/Hr
13	Turn	Medium	11.62932, 78.11659	30 KM/Hr
14	Turn	High	11.64183, 78.11810	15 KM/Hr
15	Turn	Medium	11.64338, 78.11998	30 KM/Hr
16	Turn	Medium	11.67903, 78.12796	30 KM/Hr
17	Turn	Medium	11.69532, 78.10764	30 KM/Hr
18	Turn	Medium	11.73968, 78.05098	30 KM/Hr
19	Turn	Medium	11.86790, 78.08347	30 KM/Hr
20	Turn	Medium	11.91357, 78.05868	30 KM/Hr
21	Turn	Medium	11.92556, 78.06499	30 KM/Hr
22	Turn	Medium	11.93296, 78.06432	30 KM/Hr
23	Turn	Medium	11.93535, 78.06150	30 KM/Hr
24	Turn	Medium	11.95531, 78.04698	30 KM/Hr
25	Turn	High	11.95803, 78.04964	15 KM/Hr
26	Turn	Medium	11.96068, 78.04960	30 KM/Hr
27	Turn	High	11.96370, 78.05142	15 KM/Hr
28	Turn	High	11.96542, 78.04938	15 KM/Hr
29	Turn	Medium	11.96650, 78.04927	30 KM/Hr
30	Turn	Medium	11.96890, 78.05109	30 KM/Hr
31	Turn	Medium	12.22422, 78.19127	30 KM/Hr
32	Turn	Medium	12.24685, 78.19561	30 KM/Hr
33	Turn	Medium	12.25012, 78.19946	30 KM/Hr
34	Turn	Medium	12.25280, 78.20041	30 KM/Hr
35	Turn	Medium	12.29065, 78.19892	30 KM/Hr
36	Turn	Medium	12.32012, 78.20734	30 KM/Hr
37	Turn	Medium	12.40099, 78.21805	30 KM/Hr
38	Turn	Medium	12.45136, 78.21799	30 KM/Hr
39	Turn	Medium	12.50485, 78.22336	30 KM/Hr
40	Turn	High	12.56075, 78.18183	15 KM/Hr
41	Turn	High	12.56624, 78.18435	15 KM/Hr
42	Turn	Medium	12.66016, 78.18941	30 KM/Hr
43	Turn	Medium	12.69881, 78.19213	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
44	Turn	Medium	12.70090, 78.19364	30 KM/Hr
45	Turn	High	12.70472, 78.19423	15 KM/Hr
46	Turn	Medium	12.70548, 78.19353	30 KM/Hr
47	Turn	Medium	12.70686, 78.18366	30 KM/Hr
48	Turn	Medium	12.70319, 78.17548	30 KM/Hr
49	Turn	Medium	12.70447, 78.17219	30 KM/Hr
50	Turn	High	12.70564, 78.17170	15 KM/Hr
51	Turn	Medium	12.70885, 78.16739	30 KM/Hr
52	Turn	High	12.70921, 78.16525	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
1	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
2	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
3	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
4	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
5	clinic	Dr. Dhanasekar Clinic	11.4793113, 77.8829247	30 km/h	Medium
6	hospital	Sri Vellaiyan Hospital	11.4798876, 77.8831394	30 km/h	Medium
8	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
9	hospital	Annapoorna Medical College Hospital	11.57693, 78.037627	30 km/h	Medium
10	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
11	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
12	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
14	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
19	hospital	Bharathi Hospital, Salem	11.6492647, 78.1204131	30 km/h	Medium
20	hospital	Sisu Hospital Inida Ltd	11.6505276, 78.1205003	30 km/h	Medium
21	hospital	Aishwarya Hospital	11.6716291, 78.1267807	30 km/h	Medium
23	hospital	C.M. Hospital	11.6747345, 78.1255034	30 km/h	Medium
24	pharmacy	Healthy Pharmacy	11.6762714, 78.127744	30 km/h	Medium
25	pharmacy	Thulasi Pharmacy	11.6761991, 78.1277377	30 km/h	Medium
26	hospital	Sri Gokulam Speciality Hospital	11.6788724, 78.1274258	30 km/h	Medium
27	hospital	Global Medical Center	11.6867049, 78.1186697	30 km/h	Medium
28	hospital	Global Medical Centre Jagirammapalayam	11.6894409, 78.1108847	30 km/h	Medium
29	hospital	Manipal Hospital	11.703072, 78.1026612	30 km/h	Medium
33	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
34	hospital	St. Joseph's Speciality Hospital	11.761611, 78.0522344	30 km/h	Medium
37	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
38	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
39	hospital	Thoppur,Government Hospital	11.9398824, 78.0551658	30 km/h	Medium
41	hospital	Government Hospital	12.3048002, 78.2060881	30 km/h	Medium
46	hospital	TCR Multi Speciality Hospital	12.5088183, 78.2237657	30 km/h	Medium
47	hospital	St. Mary's Hospital	12.5128555, 78.2186887	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
48	hospital	KVS Hospital	12.5277857, 78.2055049	30 km/h	Medium
49	hospital	Ga Heart Hospital	12.5287708, 78.2055336	30 km/h	Medium
50	clinic	Psychiatric Centre	12.529476, 78.205064	30 km/h	Medium
51	clinic	Shree Ramachandra ENT Centre	12.5290096, 78.2037359	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
13	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
15	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
16	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
17	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
18	school	Diamond Rays Matric HR Sec School	11.6385629, 78.1206259	30 km/h	Medium
22	college	Sona College	11.6768, 78.1257052	30 km/h	Medium
30	college	Govt Engineering College	11.7128537, 78.0877084	30 km/h	Medium
31	university	Periyar University	11.7186758, 78.0771089	30 km/h	Medium
32	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
35	school	John Britto Matriculation Higher Secondary School	11.7676907, 78.0513092	30 km/h	Medium
36	college	Dhirajlal Gandhi College of Technology	11.7968014, 78.0562095	30 km/h	Medium
40	college	Laxminarayana Polytechnic College	12.0421277, 78.103203	30 km/h	Medium